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Characterization of Al-Asfar Lake Region in Al-Hassa Oasis Using a 3D Gravity-Based Analysis

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Abstract

This study aims to investigate and reconstruct the 3D subsurface model of the Al-Hassa region, focusing on Al-Asfar Lake, a critical wetland shallow lake within the UNESCO-designated Al-Hassa Oasis. Primarily fed by agricultural drainage, Al-Asfar Lake influences the region's surface and groundwater systems. The objective is to characterize subsurface density variations and identify structural discontinuities, such as faults and fractures, impacting groundwater pathways and geological formations, providing essential insights for resource management and protection of this unique oasis and wetland ecosystem. A comprehensive gravity survey was conducted across the Al-Hassa region, specifically around Al-Asfar Lake, to map subsurface density variations and structural features influencing water flow and storage. Gravity measurements were taken on a structured grid and processed with corrections for terrain, drift, and instrumental noise to ensure data accuracy. Advanced modeling techniques generated a high-resolution 3D model, revealing lateral density contrasts and potential geological boundaries. This approach facilitates the identification of pathways and barriers to groundwater movement, yielding critical data for understanding hydrological processes that sustain Al-Asfar Lake and the broader Al-Hassa Oasis region. The gravity survey data revealed significant density variations across the study area, highlighting structural discontinuities likely associated with fault zones and fractures, which play a critical role in groundwater movement and distribution. These density contrasts identified areas of varying rock types and structural configurations, which are essential for understanding how subsurface features influence groundwater flow toward Al-Asfar Lake. The resulting 3D subsurface model offers a detailed view of hydrogeophysical dynamics, illustrating regions with varying groundwater storage potential. This model provides a valuable tool for assessing groundwater availability within the Al-Hassa Oasis ecosystem, supporting the sustainable management of Al-Asfar Lake. This study presents a novel application of gravity measurements for mapping hydrogeophysical features in Al-Hassa. By focusing on Al-Asfar Lake, the findings contribute to the

knowledge of subsurface characterization, supporting groundwater management and preservation strategies in arid and agriculturally influenced environments.

Introduction

Al-Asfar Lake, located in the eastern region of the Al-Hassa Oasis in eastern Saudi Arabia, represents one of the largest inland wetland systems in the Arabian Peninsula. The lake functions as a terminal basin receiving agricultural drainage and surface runoff and has been the focus of environmental and water quality investigations due to its ecological sensitivity and anthropogenic pressure (El Mahmoudi et al., 2008; Hussein et al., 2016). Despite its environmental significance, the lake's subsurface structure and potential hydraulic connectivity with surrounding aquifers remain poorly understood. The Al-Hassa region lies on the stable interior platform of the Arabian Plate and exhibits a stratigraphic sequence dominated by carbonate and evaporitic units, including the Umm er Radhuma, Rus, and Dammam Formations (UN-ESCWA and BGR, 2013; Khogali et al., 2024a). These formations comprise the main aquifer systems in the area, hosting significant groundwater reserves. However, their productivity and water quality are affected by structural heterogeneity, salinization, and anthropogenic stress. Recent studies have mapped large-scale density variations and aquifer boundaries using gravity, and electromagnetic methods (Khogali et al., 2024a; 2024b; 2024c), but none have specifically targeted Al-Asfar Lake as a unique hydrogeological zone.

The gravity method plays a significant role in groundwater exploration, due to its capability in mapping density distributions that correlate with groundwater basins and basement configurations, and identifying the boundary between different geological formations, which is crucial for understanding aquifer distribution (Blakely, 1995). Gravity data can effectively delineate subsurface structures and lithology, such as bedrock topography, which can assist in identifying potential groundwater reservoirs, provided there is a notable density contrast present. Previous studies have successfully applied the gravity method in various hydrogeophysical contexts, such as in distinguishing igneous from weathered rocks and identifying potential aquifers in granite rock environments (Murty and Raghavan, 2002), mapping the thickness and potential of groundwater in sedimentary rock environments (Chandler, 1994), and exploring groundwater in glacier regions by mapping buried valleys and bedrock depth (Carmichael and Henry, 1977). Moreover, changes in groundwater storage can be detected through gravity anomaly differences measured at different times, which can be applied in aquifer monitoring and water management to detect adverse changes such as excessive pumping and reduced recharge (Gehman et al., 2009; Kennedy et al., 2014; Sendrós et al., 2023). Traditionally used for regional surveys, gravity data now allow for the investigation of small density changes caused by decreasing water level within unconfined aquifers (Kennedy et al., 2014; González-Quirós and Fernández-Álvarez, 2014), while seasonal hydrological changes can also be identified by applying the micro-gravity method (Kang et al., 2011; Tanaka et al., 2013). For broader regional environmental and groundwater research, satellite gravity data from the GRACE (Gravity Recovery and Climate Experiment) mission are used to estimate the dynamic changes and trends in water mass stored as groundwater across large areas, offering insights that ground-based methods alone cannot provide, although with lower resolution (Chen et al., 2016; Mohamed et al., 2017; Rodell et al., 2018). In Saudi Arabia, GRACE data have been used to quantify groundwater recharge and depletion resulting from climatic and human activities (e.g., Fallatah et al., 2019; Mohamed et al., 2022; Alshehri and Mohamed, 2023). Despite the non-uniqueness of the gravity data analysis results and the effect of external (not related to geology) sources on the data, the gravity method offers the advantages of being reasonably quick, inexpensive, and non-disruptive (Telford et al., 1990; Blakely, 1995; Fairhead, 2016). To enhance its effectiveness in delineating aquifer boundaries and structural features, the gravity method is usually applied in conjunction with other geophysical methods, such as the electric resistivity and time-domain electromagnetic, while additional geoscientific tools (geological mapping and cross sections, drilled wells and logs, petrophysical investigations, etc.) can provide constraints to the interpretation of the results.

The present study focuses on the Al-Asfar Lake region, situated within the Al-Hassa Oasis. This work aims to apply high-resolution 3D gravity modeling to characterize the subsurface structures that potentially influence local hydrogeological behavior. By enhancing the structural understanding of this unique environment, the results will contribute to the broader goal of sustainable groundwater management and environmental monitoring in the oasis system.

Geology and hydrogeology of the study area

The Al-Hassa region, located in eastern Saudi Arabia, lies within the Arabian Shelf, a tectonically stable platform overlying the Precambrian crystalline rocks of the Arabian Shield. This gently subsiding shelf forms part of the Arabian Plate's eastern margin and hosts a thick sedimentary succession deposited from the Paleozoic to the Quaternary, making it one of the most geologically and hydrogeologically important regions of the Arabian Peninsula (UN-ESCWA and BGR, 2013). The stratigraphic column is dominated by Paleogene carbonate formations, particularly the Umm er Radhuma and Dammam formations—overlain by the Rus Formation, which is rich in gypsum and anhydrite and acts as a regional aquitard (Al Tokhais and Rausch, 2008). More recent Neogene–Quaternary deposits unconformably overlie the older sequences and consist of unconsolidated alluvial, aeolian, and sabkha sediments. These shallow surface deposits constitute the uppermost hydrostratigraphic units and play a crucial role in the recharge, salinity, and vulnerability of shallow aquifers around Al-Asfar Lake.

At the local scale, the surface geology of the Al-Asfar area is characterized by three dominant lithofacies: eolian sands (Qe), gravel deposits (Qg), and sabkha and khabra deposits (Qsk). These are illustrated in Fig. 1, which shows the spatial distribution of sediment types alongside the gravity survey stations used in recent geophysical studies. The sabkha plains (Qsk) dominate the southwestern and southeastern margins of the lake and are associated with high salinity and low permeability. In contrast, the northern sector features gravel-dominated terrain (Qg), indicative of alluvial fan activity and higher infiltration potential, while eolian sands (Qe) dominate the eastern and southern sectors (Mohammed et al., 2022).

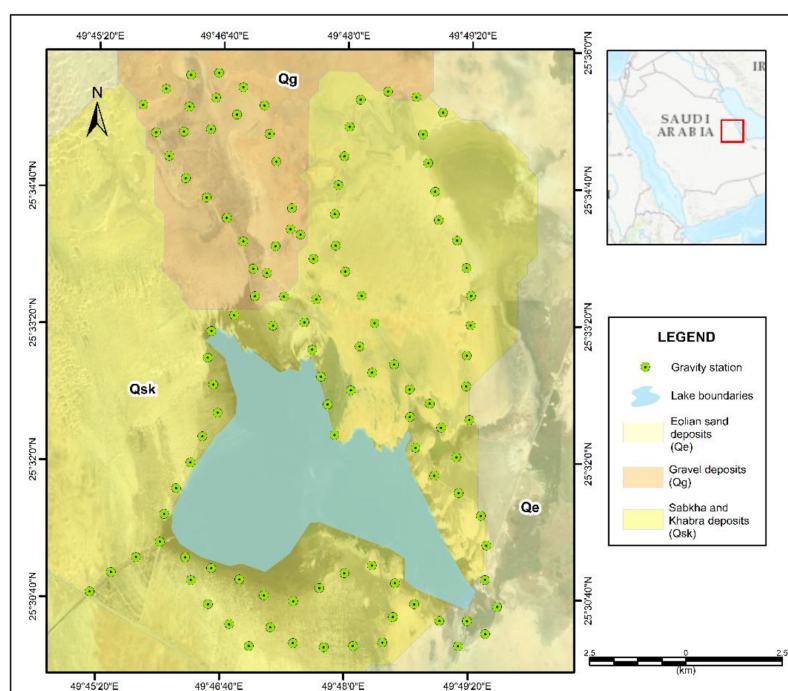


Figure 1—Map of the Al-Asfar Lake area, highlighting the surface geologic formations and the expanse of the lake. The green points indicate the locations of the acquired gravity stations.

The structural setting of the region is defined by its location on the eastern limb of the Ghawar Anticline, a N-S trending megastructure that is globally recognized as the largest onshore hydrocarbon reservoir. The anticline trends N–S to NNE–SSW, with bedding planes dipping gently 1–3° eastward, consistent with regional basin ward tilting. This structure has significantly influenced subsurface fluid flow, not only in petroleum systems but also in groundwater circulation within the underlying carbonate aquifers (Khogali et al., 2024b). The regional tectonic stress regime is dominated by WNW–ESE compression, inherited from the Late Cretaceous–Miocene convergence of the Arabian and Eurasian Plates. This has led to the development of asymmetric folds and associated normal and strike-slip faults, some of which have been later reactivated or altered by salt tectonics, particularly involving the mobile evaporites of the Rus Formation. Geophysical studies have identified subsurface gravity anomalies, interpreted as buried salt domes and collapse structures, especially in areas surrounding Al-Asfar Lake (Khogali et al., 2024a; 2024b). Further evidence of tectonic and halokinetic influence is reflected in the orientation of fracture corridors, predominantly trending NNW–SSE and NE–SW. These features likely act as preferential pathways or barriers for groundwater movement, affecting both the hydraulic connectivity between aquifer layers and the spatial variability of salinity. Although the region is not considered microseismicity active, minor micro seismicity has been detected offshore in the Arabian Gulf, suggesting deep structural adjustments are ongoing, though the current tectonic regime is passive (UN-ESCWA and BGR, 2013).

Al-Asfar Lake is part of a broader hydrological system fed primarily by shallow groundwater and agricultural drainage. The main aquifer systems in the area are the Umm er Radhuma, Dammam, and Neogene aquifers (UN-ESCWA and BGR, 2013; Al Tokhais and Rausch, 2008). The Umm er Radhuma aquifer is a deep, regionally extensive karstified carbonate unit that provides water for both industrial and domestic uses. Recharge is limited and occurs mainly through outcrops and indirect infiltration processes (UN-ESCWA & BGR, 2013). The Dammam aquifer, overlying the Umm er Radhuma, is a semi-confined unit composed of dolomitic limestone and marl. It is commonly exploited for irrigation and is more vulnerable to salinization due to over-pumping and reduced recharge (El Mahmoudi et al., 2011; Al-Omran et al., 2018). The Neogene aquifer represents a shallow, unconfined system that has been heavily exploited, particularly for agricultural purposes. Its productivity is variable, with low recharge and signs of degradation linked to overuse and contamination (Al Tokhais and Rausch, 2008; Mohammed et al., 2022).

Groundwater geochemistry studies indicate that most samples fall within the Na–Cl and Ca–Mg–SO₄ facies, with total dissolved solids (TDS) often exceeding drinking water standards. Salinity and hardness vary significantly, with several locations classified as "poor" or "very poor" for drinking purposes according to water quality index (WQI) assessments (Al-Omran et al., 2018). Recent spatial analyses by using data-driven GIS techniques identified groundwater potential (GWP) zones in the Al-Hassa Oasis. Soil texture and geology were found to be the most influential factors for groundwater availability, while fracture density and topographic indices had lower influences. These results support the need for integrated hydrogeological assessments when considering sustainable groundwater extraction strategies. Moreover, evapotranspiration (ET) assessments using SEBAL and Landsat-8 imagery showed that actual ET rates over Al-Asfar Lake and surrounding agricultural lands ranged from 800 to 2000 mm/year, reinforcing the significant water loss through surface processes in this arid environment (Biro et al., 2019).

Data and Methodology

A gravity survey comprising 111 relative gravity stations was conducted in the Al-Asfar Lake region during April 2025. The measurements, arranged in a grid pattern with 500-meter spacing, covered approximately 60 km² and were adapted to suit the terrain and accessibility constraints. A CG-5 AUTOGRAV™ gravimeter (Scintrex Ltd.) was used during acquisition, with built-in mechanisms for filtering out seismic disturbances and spikes, as well as correcting for Earth tides and instrument tilt. Gravity data acquisition is inherently vulnerable to various external factors, including anthropogenic noise (e.g., traffic, machinery) and natural

influences (e.g., wind, temperature shifts, seismic activity) (Blakely, 1995). To obtain reliable data, rigorous quality control and corrections were applied to mitigate these effects. Each station underwent three 60-second readings, with additional measurements taken if discrepancies exceeded 10 μGal , tilt surpassed ± 10 arcsec, or if environmental noise was detected (e.g., wind gusts or nearby vehicles).

Due to the absence of a known absolute gravity reference in the area, the base station's theoretical gravity value served as the standard for all relative gravity readings. Should absolute reference become available, the data could be tied to the International Absolute Gravity Base Station Network (IAGBN). A Trimble R10 Differential GPS was employed to determine precise station coordinates and elevations, ensuring horizontal and vertical accuracy within 1 cm. Data processing was carried out in Seequent's Oasis montaj platform, utilizing its Gravity and Terrain Correction module. The initial data adjustments included corrections for Earth tides, instrument height, and instrument drift. Tidal corrections, influenced primarily by lunar and solar gravitational forces, depend on geographic position and measurement timing (Amarante and Trabanco, 2016). The theoretical gravity at the base station was calculated using the 1980 International Gravity Formula (Moritz, 1980), based on latitude alone, forming the baseline for correcting all other measurements. Instrumental drift for each day was accounted for using a linear interpolation between morning and evening base station readings. Further corrections included adjustments for latitude effects and local topography. Topographic influences were modeled using two Shuttle Radar Topography Mission (SRTM) digital elevation models: a 30 m resolution DEM for local detail and a 90 m resolution grid for regional context. However, no bathymetric data was available for the lake area. A standard reduction density of 2.67 g/cm^3 was applied.

The final correction stages included the application of free-air and Bouguer corrections. A standard free-air gradient of 0.3086 mGal/m (Sheriff, 1991) was used to compensate for elevation-induced gravity changes. The Bouguer correction, accounting for both terrain elevation and density between the station and sea level, was also applied using the same density value of 2.67 g/cm^3 . Following all corrections, the data were interpolated using the minimum curvature technique to generate the complete Bouguer anomaly map (Fig. 2a), which incorporates both deep and shallow subsurface effects (Fairhead, 2016). To differentiate these effects, a first-order polynomial surface was subtracted from the Bouguer map, separating regional gravity trends from localized anomalies (Telford et al., 1990) (Fig. 2b). Processed gravity data were further enhanced using various filters. One effective approach involves calculating gravity field derivatives to highlight structural boundaries. Among these, the tilt derivative—a ratio of vertical to total horizontal gradients—is commonly used due to its ability to highlight subtle anomalies and maintain low noise sensitivity (Miller and Singh, 1994). Positive tilt values correspond to anomaly sources, while zero contours delineate significant structural features like faults, lithological contacts, or salt boundaries. In this study, tilt derivatives were derived from the residual gravity map, and zero-value contours were manually traced to produce a lineament interpretation (Fig. 3).

A three-dimensional (3D) inversion of the residual gravity field was carried out using the VOXI Earth Modeling tool within Oasis montaj. VOXI relies on the Cartesian cut cell method (Ingram et al., 2003), modeling the subsurface as a grid of prismatic voxels. Algorithm improvements by Ellis and MacLeod (2013) allowed more accurate modeling of geological boundaries. To minimize boundary effects, the model domain was expanded in all spatial dimensions, and grid parameters such as padding and expansion ratios were optimized. A fitting tolerance of 5% of the data's standard deviation was used. The most reliable results were observed in the central zone beneath Al-Asfar Lake, with voxel size increasing with distance from this core area. The final unconstrained inversion used a 3D grid composed of $29 \times 42 \times 14$ voxels (Easting $\times$ Northing $\times$ Depth), with a model resolution of 250 meters and a maximum investigation depth of 2.9 km.

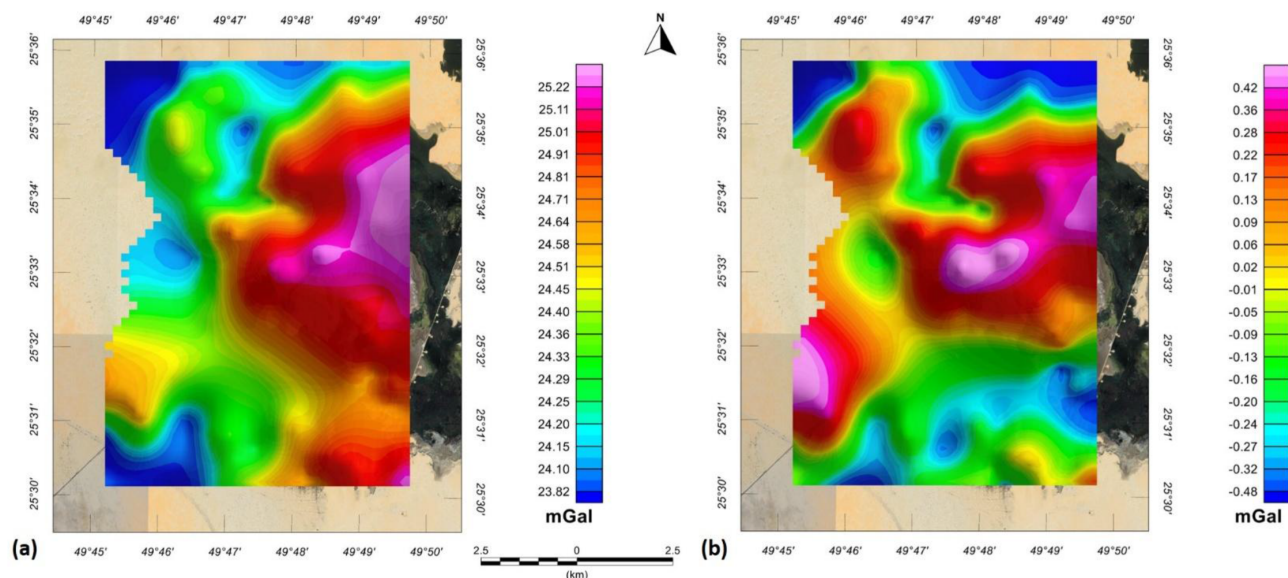


Figure 2—(a) Complete Bouguer, and (b) residual gravity field maps of the study area.

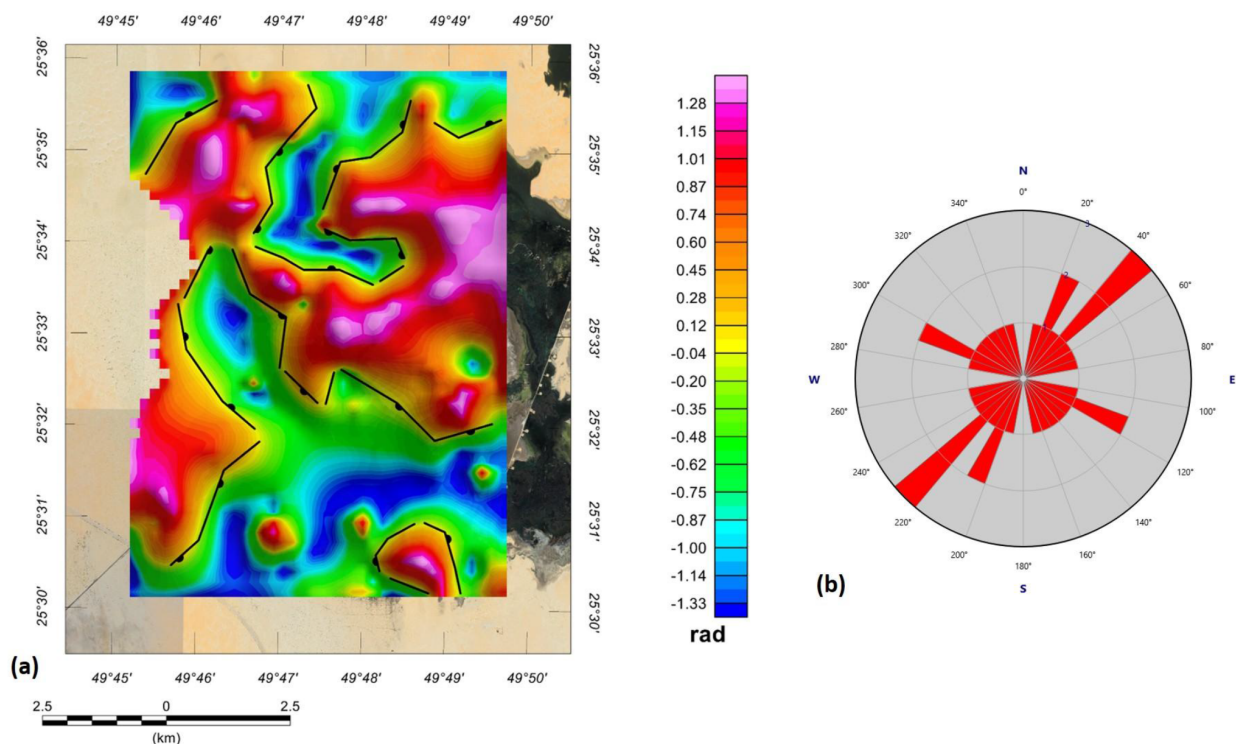


Figure 3—(a) Tilt derivative map of the study area. The interpreted lineaments that correspond to geological and structural edges are indicated as black lines, with the semi-circles showing their dipping direction; (b) Rose diagram of the interpreted lineaments.

Results

Fig. 2 shows the gravity field variations as calculated by the analysis of the acquired field data. More specifically, Fig. 2a is the map of the complete Bouguer anomaly, which represents the influence of long- and short-wavelength anomalies. Thus, it contains the signature of both deep (regional field) and shallow (residual field) structures in the gravity field. The eastern part of the study area is dominated by high-gravity values, while lower values are observed in the northern and southern parts. The residual gravity field map, which is an indication of the effect of near-surface structures, is presented in Fig. 2b. The larger part of the

study area is governed by high gravity values, with the exception of the southern part, where lower values are observed. The low values in the northern part, although they might indicate a change in the geotectonic regime, should not be considered for the interpretation, as they are located at the edge of the survey grid.

The tilt derivative map is presented in Fig. 3a. Similarly to the gravity field maps, the eastern part of the study area is mostly characterized by positive tilt angle values, while negative values are observed in the southern part. The black lines depict the manually interpreted lineaments, which possibly correspond to rock contacts and/or structural and tectonic features such as faults. The orientation of the semi-circles in the lineaments indicates the dip direction of the latter, pointing towards negative tilt values. The rose diagram of the interpreted lineaments (Fig. 3b) was constructed using the GeoRose software and shows two observable trends: one trending NE-SW to NNE-SSW and another from NW-SE to NNW-SSE.

Using 3D inversion on the gravity data, a 3D model of the subsurface structures was created. Fig. 4 illustrates the negative density contrast bodies (lower than -0.002 g/cm^3 in comparison to the average Earth density of 2.67 g/cm^3 that was used), which are mainly present in the southern part of the study area, while they are also encountered in the northern and western parts, extending to depths of over 2 km. The surface expanse of the Al-Asfar Lake is depicted with a closed black line in the Figure.

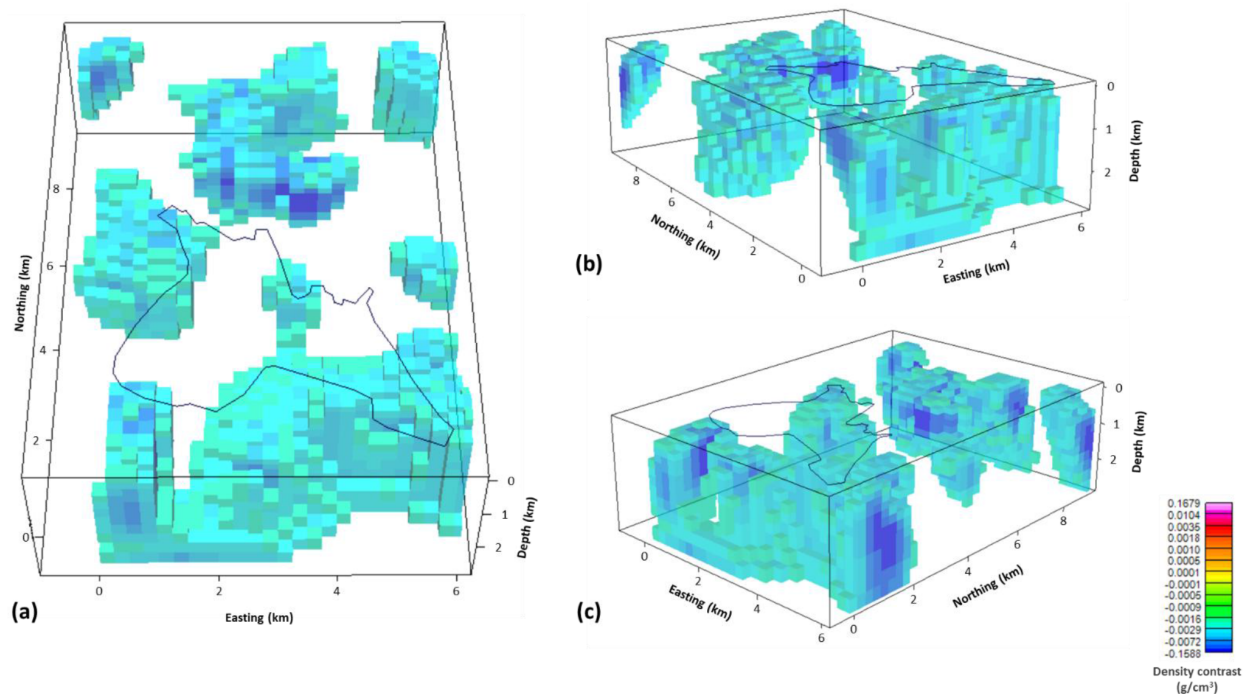


Figure 4—(a, b, c) 3D model of the study area in different side views, indicating the low-density formations (blue-colored bodies). The enclosed area within the black line corresponds to the Al-Asfar Lake expanse.

Discussion

The interpretation of the residual gravity field map and 3D inversion model over the Al-Asfar Lake region offers significant insights into the subsurface geological and hydrogeological architecture, reinforcing previous structural and stratigraphic models while introducing new evidence for groundwater dynamics. A pronounced low-gravity anomaly detected in the southern part of Al-Asfar Lake, interpreted as a low-density zone in the 3D inversion model, is a particularly important feature. These low-density bodies are spatially extensive and could correspond to unconsolidated sediments, partially saturated or variably compacted alluvium, or even dissolution-affected formations. Their location and density contrast suggest that they may serve as vertical or lateral pathways for groundwater movement from deeper aquifers toward the surface. This interpretation aligns with the hydrogeological setting of the region. Al-Asfar Lake is situated

in a topographic and hydrologic low, functioning as a terminal basin for agricultural runoff but potentially also as a discharge point for shallow or even intermediate-depth groundwater systems. The main aquifers in the area, the Umm er Radhuma and Dammam formations, consist of Paleogene carbonate rocks with significant potential for water storage and transmission, particularly where karstification and fracturing occur (UN-ESCWA and BGR, 2013; Al Tokhais and Rausch, 2008). These aquifers dip gently toward the east, consistent with the structural tilt of the Arabian Shelf, and are overlain by the Rus Formation, a semi-permeable evaporitic unit that can act as a leaky confining layer depending on local heterogeneities. The low-density structures in the western part of the lake, as identified in the residual gravity field map, exhibit similar characteristics and may perform an analogous hydrogeological function. However, these features might also correspond to deep evaporitic or clay-rich units, as suggested by Khogali et al. (2024a; 2024b), which could act as hydraulic barriers or seals, particularly in the context of the broader Ghawar Anticline system. These evaporitic bodies may originate from older structural lows or dissolution collapse zones and play a role in creating localized aquitards, affecting the upward or lateral movement of groundwater.

In contrast, the northern part of the lake is associated with higher gravity values, indicative of denser formations. These may correspond to more compacted lithologies or regions where erosion has exposed older, consolidated units. A small low-density anomaly located at the northeastern margin remains ambiguous due to its peripheral location and potential edge effects in the inversion model, limiting the confidence in its interpretation. One of the most striking structural observations from geophysical modeling is the alignment between identified surface/subsurface lineaments and the edges of the low-density bodies. The detected NNW–SSE and NNE–SSW oriented fractures or faults are structurally consistent with regional tectonic trends reported in Al-Hassa and adjacent areas (Saner et al., 2005; Khogali et al., 2024a; 2024b). These features likely represent inherited basement structures or halo kinetically induced fractures that have been reactivated during later extensional or gravitational collapse phases. Their orientation aligns well with the regional stress regime, which trends WNW–ESE, promoting the development of oblique fault systems. These tectonic elements are critical in the hydrogeological functioning of the lake system. Faults and fracture corridors can enhance secondary permeability, enabling localized vertical connectivity between aquifers and the lake. This would help explain not only the sustained water levels in Al-Asfar Lake, despite high evaporation rates, but also the heterogeneous distribution of soil moisture and vegetation in the surrounding areas, as noted in remote sensing observations (Hassaballa and Salih, 2024). However, the lack of direct subsurface verification—such as borehole data, geophysical imaging from within the lake, or in situ hydraulic testing—limits the ability to confirm active recharge from these structures. Interestingly, the E–W oriented lineaments identified in previous regional studies (Khogali et al., 2024b) were not near Al-Asfar Lake. This suggests that the structural regime in the lake area is more segmented or influenced by localized deformation, potentially related to salt movement or differential compaction rather than regional tectonic controls alone. The absence of E–W features may also reflect a lack of data resolution or contrasts in rock properties that limit their detection via gravity inversion.

The integration of geological, geophysical, and hydrogeological data thus reveals a complex but coherent picture. Al-Asfar Lake appears to be situated in a structurally controlled depression, influenced by both regional tilting and localized fracturing. The low-density zones detected using geophysics are consistent with lithologies that can serve as potential conduits for groundwater discharge, while lineament orientations support a tectonically modulated permeability framework. These findings reinforce the idea that the lake's hydrology cannot be attributed solely to surface processes (e.g., agricultural drainage), but instead it is likely governed by a multi-layered interaction between lithology, structure, and subsurface flow dynamics.

Further investigations should focus on quantifying the petrophysical properties of the low-density formations—especially their porosity, permeability, and connectivity—to evaluate their function as effective water carriers. Additionally, direct hydrogeological surveys within the lake and surrounding units are needed to verify the presence of recharge pathways, evaluate water balances, and model flow directions, which

are currently inferred rather than observed. These steps are essential for building a robust water resource management strategy in this environmentally sensitive and hydrologically complex system.

Conclusions

The results of this study highlight the importance of combining geophysical models with geological and hydrogeological context to infer hidden water dynamics in desert oases. These insights are not only valuable for understanding local processes but also for developing regional water management strategies. Ongoing studies are currently focusing on validating these interpretations through direct field measurements and expanding the structural mapping across adjacent zones of the Al-Asfar Lake region.

An expansion of the gravity survey in the northern portion of the study area might provide further insights into the regional geotectonic regime associated with the observed low-gravity anomalies. Such an effort could help refine the current 3D model by delineating the exact geometry and spatial extent of the low-density bodies in the northern sector of Al-Asfar Lake. Moreover, integrating the newly acquired gravity data with those from previous studies in the Al-Hassa region (Khogali et al., 2024a, (2024b) may reveal broader structural patterns and improve the understanding of fault connectivity, salt tectonics, and their influence on subsurface fluid flow.

Additionally, coupling gravity data with other geophysical methods—such as electrical resistivity tomography (ERT), seismic refraction, or electromagnetic induction—would allow for more robust interpretation of the physical properties (e.g., porosity, lithology, and saturation) of the identified structures. This integrated approach could validate whether the low-density anomalies function as active groundwater pathways or are instead related to lithological variations without hydraulic connectivity. Given the role of these structures in potentially sustaining lake levels in an arid environment, understanding their petrophysical and hydrogeological behavior is critical. Future research should include in situ investigations—such as shallow drilling, downhole logging, and aquifer pumping tests—to verify recharge mechanisms and quantify groundwater inflow rates to the lake.

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